

AP CALCULUS AB - UNIT 5

Guided Notes 5.2–5.5 I: Derivative Graph Analysis - Part 1



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we find critical numbers and intervals of increase and decrease?

- Find critical numbers and intervals of increase and decrease.
- Use the First Derivative Test to classify local extrema.
- Apply the Extreme Value Theorem and closed-interval method for absolute extrema.
- Connect the sign of f' with the shape of f .

Key Knowledge, Vocabulary, and Notation

Critical numberAn input c in the domain of f where $f'(c) = 0$ or $f'(c)$ does not exist.**Increasing/decreasing** $f' > 0$ gives increasing; $f' < 0$ gives decreasing on an interval.**First Derivative Test** $+$ $\rightarrow$ $-$ gives a local maximum; $- \rightarrow +$ gives a local minimum.**Extreme Value Theorem**

A continuous function on a closed interval attains absolute maximum and minimum values.

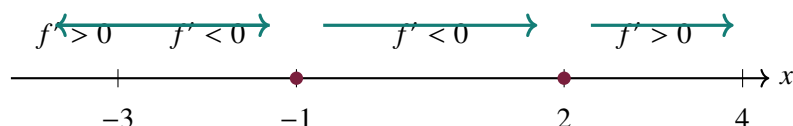
Closed-interval methodEvaluate f at endpoints and all interior critical numbers; compare the values.**Candidate distinction**

A critical number is only a candidate; classification requires sign change or another valid test.

Concept Development and Strategy

1. Find the domain first, then solve $f' = 0$ and include points in the domain where f' is undefined.
2. Build a sign chart for f' using test inputs or factor signs.
3. State extrema as points or values as requested, and distinguish local from absolute.
4. For an absolute-extrema problem on $[a, b]$, endpoints are mandatory candidates even though derivatives are not required there.
5. When given the graph of f' , local extrema of f occur at sign changes, not merely at zeros of f' .

Shared Representation / Data



Worked Example: First derivative sign chart

For $f(x) = x^3 - 3x^2 - 9x + 1$, find intervals of increase/decrease and local extrema.

Answer and Reasoning

$f' = 3(x - 3)(x + 1)$. Increasing on $(-\infty, -1) \cup (3, \infty)$, decreasing on $(-1, 3)$. Local max at $x = -1$ with $f(-1) = 6$; local min at $x = 3$ with $f(3) = -26$.

Worked Example: Critical derivative undefined

Analyze $f(x) = x^{2/3}$ near 0.

Answer and Reasoning

$f' = \frac{2}{3}x^{-1/3}$ is undefined at 0, which is in the domain. The derivative changes from negative to positive, so f has a local and absolute minimum at $(0, 0)$.

Worked Example: Closed interval

Find absolute extrema of $f(x) = x^3 - 3x$ on $[-2, 2]$.

Answer and Reasoning

Critical numbers $x = \pm 1$. Values: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(2) = 2$. Absolute max 2 at $x = -1, 2$; absolute min -2 at $x = -2, 1$.

Worked Example: Graph of f'

Using the shared sign chart, classify the critical numbers.

Answer and Reasoning

At $x = -1$, f' changes $+$ $\rightarrow$ $-$, so f has a local maximum. At $x = 2$, f' changes $-$ $\rightarrow$ $+$, so f has a local minimum.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. Find critical numbers of $f(x) = x^2 - 6x + 1$.

2. If $f' > 0$ on $(1, 5)$, describe f there.

3. For $f' = x(x - 4)$, determine intervals where f increases.

4. Classify $x = 0$ and $x = 4$ from the previous derivative.

5. For $f(x) = x + 4/x$ on $(0, \infty)$, find the absolute minimum.

6. Find absolute extrema of $f(x) = x^{2/3}(x - 4)$ on $[-1, 8]$.

7. Construct a differentiable function with a critical point at 0 that is neither maximum nor minimum.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. A critical number must

- (A) make $f' = 0$ only
(B) be in the domain of f and make $f' = 0$ or undefined
(C) be an endpoint
(D) be a local extremum

2. If f' changes from negative to positive at c , then f has

- (A) a local maximum
(B) a local minimum
(C) an inflection point necessarily
(D) no conclusion

3. For $f(x) = x^3 - 6x^2 + 9x$ on $[0, 5]$, the absolute maximum occurs at

- (A) $x = 0$
(B) $x = 1$
(C) $x = 3$
(D) $x = 5$

Common Errors and AP Precision

- A zero derivative or zero second derivative is a candidate condition, not by itself a complete extremum or inflection justification.
- State theorem hypotheses, include endpoints in absolute-extrema work, and verify every optimization candidate on the feasible domain.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

Prove that a differentiable function with $f' > 0$ everywhere on an interval has at most one zero there.

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